Notes: Acemoglu & Restrepo (QJE, 2026)

Automation and Rent Dissipation: Implications for Wages, Inequality, and Productivity

Contents

1 Big picture **3**

2 Data and measurement **3**
2.1 Demographic groups and wage outcomes . . . 3
2.2 Task displacement from automation . . . 4
2.3 Rent and job-quality measures . . . 4
2.4 Aggregate data for the quantitative exercise . . . 4

3 Models and empirical specifications **5**
3.1 Task production . . . 5
3.2 Worker rents as wage wedges . . . 5
3.3 Equilibrium representation . . . 5
3.4 Inefficiency from rents . . . 6
3.5 Endogenous targeting of high-rent tasks . . . 6
3.6 Within-group wage compression . . . 6
3.7 Wage effects without ripple effects . . . 6
3.8 TFP effects . . . 6
3.9 Reduced-form wage specification . . . 7
3.10 Rent-shift specification . . . 7
3.11 General equilibrium estimation . . . 7

4 Identification and empirical design **7**
4.1 What identifies the reduced-form relationships . . . 7
4.2 Key identifying assumption . . . 8
4.3 Separating rents from compensating differentials . . . 8
4.4 Why the U-shape is important . . . 8
4.5 Limits and threats . . . 8

5 Tables and figures **9**
5.1 Figure I and Figure II: theory of wage compression and ripple effects . . . 9
5.2 Figure III and Figure IV: measuring exposure and average wage effects . . . 10
5.3 Figure V and Figure VI: the U-shape in within-group wage changes . . . 11
5.4 Figure VII: decomposing base wages and rents . . . 11
5.5 Figure VIII and Figure IX: direct rent proxies . . . 12
5.6 Table I: general equilibrium quantitative effects . . . 13
5.7 Figure X and Figure XI: model decomposition and fit . . . 14

6 Short summary **16**

7 Conclusion **17**
7.1 Core conclusion . . . 17
7.2 Economic intuition . . . 17

7.3 Labor-market interpretation . . . 17
7.4 Inequality implications . . . 17
7.5 Productivity and welfare implications . . . 17
7.6 Policy implications . . . 18
7.7 Managerial and firm-level implications . . . 18
7.8 Research agenda . . . 18
7.9 Final takeaway . . . 18

1 Big picture

- **Question.** How does automation affect wages, inequality, productivity, and welfare when some jobs pay worker rents, meaning wages above outside options or opportunity costs?
- **Core mechanism.** If a task pays a high rent, the wage paid by the firm is high relative to the worker's opportunity cost. Firms therefore have an especially strong private incentive to automate high-rent tasks.
- **Rent dissipation.** Automation shifts workers out of high-rent tasks and into lower-rent jobs. This dissipates rents that workers previously received.
- **Wage implication.** In a competitive task model, automation lowers the relative demand for exposed groups. With rents, the wage loss is larger because exposed workers also lose the high-rent jobs that used to lift their earnings.
- **Within-group inequality implication.** Automation can *reduce* within-group wage dispersion for exposed groups. The reason is that automation hits the upper part of the within-group wage distribution, where rents are concentrated, more than the lower part.
- **Productivity implication.** In a frictionless model, automation raises productivity by reducing costs. With rents, the same adoption decision can be inefficient because firms automate jobs whose private cost is high due to rents rather than jobs whose social opportunity cost is high.
- **Empirical setting.** The paper studies 500 U.S. demographic groups from 1980 to 2016. Groups are defined by education, gender, age, race/ethnicity, and native/immigrant status.
- **Main reduced-form evidence.** Groups more exposed to automation experience larger average wage declines and a U-shaped pattern of within-group wage changes: wage losses are more severe around upper-middle percentiles than at the bottom.
- **Rent evidence.** Using wage differentials, displacement losses, and quit rates as rent proxies, the paper finds that automation moves exposed workers away from high-rent jobs.
- **Quantitative result.** Automation explains about 52% of the rise in between-group wage inequality since 1980. About 42 percentage points come from standard labor-demand effects; about 10 percentage points come from rent dissipation.
- **Aggregate result.** Automation's direct cost-saving effect raises TFP by about 3%, but inefficient rent dissipation offsets much of this gain. The baseline net contribution to TFP is only about 0.3%, with robustness estimates between approximately 0.3% and 1.3%.
- **Takeaway.** The private profitability of automating a job can exceed the social value of automating it when wages contain rents. Automation can therefore look productivity-enhancing to firms while reducing rents, amplifying inequality, and weakening aggregate productivity gains.

2 Data and measurement

2.1 Demographic groups and wage outcomes

- The empirical unit is a demographic group g.
- The paper uses 500 groups defined by five education levels, gender, five age groups, five race/ethnicity categories, and native/immigrant status.
- Wage changes are measured from the 1980 U.S. Census and pooled 2015–2017 American Community Survey data.
- Main outcomes include average real hourly wages and within-group wage percentiles, $\Delta \ln w_g(p)$ for $p = 5, 10, \ldots, 95, 99$.

Interpretation: The paper does not compare workers one by one. It compares groups with different baseline exposure to tasks that later became automatable. This lets the authors study both between-group inequality and within-group compression.

2.2 Task displacement from automation

The task-displacement measure is the key exposure variable. It follows the logic of Acemoglu and Restrepo's task framework and is constructed from industry exposure, routine-task specialization, and industry labor-share declines attributed to automation:

$$\text{Task displacement}_g = \sum_i \frac{\ell_{gi}}{\ell_g} RCA_{gi}^{routine} \times \frac{-\Delta \ln s_{\ell i}^A}{1 + s_{\ell i}(\lambda - 1)\pi_i} \times \frac{\mu_i}{\mu_i^A}. \tag{1}$$

- ℓ_{gi}/ℓ_g is group g's baseline employment share in industry i.
- $RCA_{gi}^{routine}$ measures how specialized group g is in routine jobs within industry i.
- $-\Delta \ln s_{\ell i}^A$ is the labor-share decline in industry i attributed to automation.
- The denominator adjusts the labor-share decline for substitution and rent effects.
- The industry automation component uses three proxies: industrial robots, specialized software services, and dedicated machinery.

Interpretation: The measure asks: given where a group worked in 1980 and which tasks it performed, what share of its tasks were exposed to the kinds of automation that reduced labor shares across industries?

2.3 Rent and job-quality measures

The paper uses several proxies for rents:

- industry and occupation wage differentials in 1980;
- wage losses after displacement from the CPS Displaced Worker Supplement;
- employment-to-employment quit rates and voluntary employment-to-unemployment quit rates, used as inverse measures of rents;
- job-quality measures from the 1977 Quality of Employment Survey to separate rents from compensating differentials.

Interpretation: The rent proxies are deliberately diverse. Wage differentials can be contaminated by productivity or amenities, while displacement losses and quit rates more directly capture job-specific rents. Similar results across proxies strengthen the rent-dissipation interpretation.

2.4 Aggregate data for the quantitative exercise

- Industry data come from the BEA Integrated Industry-Level Production Accounts.
- GDP and capital measures come from the BLS.
- Consumption is from the BEA.
- TFP is taken from Fernald's data.
- The quantitative exercise uses initial factor shares, industry shares, task-displacement estimates, cost savings from automation, and the estimated rate of rent dissipation.

Interpretation: The reduced-form section establishes the empirical mechanism. The general equilibrium section then asks how large the mechanism is for wages, inequality, consumption, and TFP.

3 Models and empirical specifications

3.1 Task production

A final good is produced by combining a continuum of tasks $x \in T$ with elasticity parameter $\lambda \in (0, 1)$:

$$y = \left[\frac{1}{M} \int_T (My_x)^{\frac{\lambda-1}{\lambda}} dx \right]^{\frac{\lambda}{\lambda-1}}. \tag{2}$$

Task output can be produced by task-specific capital or by workers from different groups:

$$y_x = \psi_{kx} k_x + \sum_g \psi_{gx} \ell_{gx}. \tag{3}$$

Interpretation: The model is a standard task framework, but with the important addition that labor costs can differ from workers' opportunity costs because some tasks pay rents.

3.2 Worker rents as wage wedges

The wage paid to workers from group g in task x is

$$w_{gx} = w_g \mu_{gx}, \qquad \mu_{gx} \geq 1, \tag{4}$$

where w_g is the base wage and μ_{gx} is the task-specific rent wedge.

Interpretation: The wedge can represent efficiency wages, bargaining rents, licensing, union contracts, norms, or other institutional sources of above-opportunity-cost wages. The key point is that the firm pays $w_g \mu_{gx}$, while the social opportunity cost is closer to w_g.

3.3 Equilibrium representation

Define task shares for group g and capital:

$$\Theta_g(w) = \frac{1}{M} \int_{T_g(w)} \psi_{gx}^{\lambda-1} \mu_{gx}^{-\lambda} dx, \qquad \Theta_k(w) = \frac{1}{M} \int_{T_k(w)} (\psi_{kx} q_x)^{\lambda-1} dx. \tag{5}$$

Define the average group rent:

$$\mu_g(w) = \frac{1}{M\Theta_g(w)} \int_{T_g(w)} \psi_{gx}^{\lambda-1} \mu_{gx}^{1-\lambda} dx. \tag{6}$$

The equilibrium base wage and price-index conditions are

$$w_g = \left(\frac{y}{\ell_g} \right)^{1/\lambda} \Theta_g(w)^{1/\lambda}, \tag{7}$$

$$1 = \left[\Theta_k(w) + \sum_g \Theta_g(w) \mu_g(w) w_g^{1-\lambda} \right]^{1/(1-\lambda)}. \tag{8}$$

Interpretation: The economy behaves like a CES system in which shares are endogenous task shares. Automation changes the task shares; rents change which tasks are cheap or expensive for firms.

3.4 Inefficiency from rents

Worker rents distort both the intensive and extensive margins. On the extensive margin, tasks are inefficiently automated when

$$\frac{w_g \mu_{gx}}{\psi_{gx}} > \frac{1}{\psi_{kx} q_x} > \frac{w_g \mu_g}{\psi_{gx}}. \tag{9}$$

Interpretation: The firm automates because the wage bill is high, but the planner compares capital cost to workers' true opportunity cost. If the private wage is high only because of rents, the firm may automate a task that society would prefer to keep with labor.

3.5 Endogenous targeting of high-rent tasks

New automation technologies make tasks in A_g^T feasible to automate. Firms choose to automate tasks in

$$A_g = \left\{ x \in A_g^T : \frac{w_g \mu_{gx}}{\psi_{gx}} \geq \frac{1}{\tilde{q}_x \psi_{kx}} \right\}. \tag{10}$$

Interpretation: High-rent tasks are more likely to be automated because replacing them saves a higher wage bill. This is the private adoption incentive behind rent dissipation.

3.6 Within-group wage compression

The model predicts a U-shaped change in the within-group wage distribution. If M_g is the share of no-rent tasks, then

$$\Delta \ln w_g(p) = \Delta \ln w_g \quad \text{for } p \in [0, M_g], \tag{11}$$

$$\Delta \ln w_g(p) < \Delta \ln w_g \quad \text{for } p \in (M_g, 1), \tag{12}$$

and the top percentile can return toward the base-wage change if some highest-rent tasks are not automatable.

Interpretation: Low within-group percentiles are close to base-wage jobs, while upper-middle percentiles contain rent-paying jobs that are newly automatable. That is why the model predicts wage declines that are most negative in the upper-middle of the within-group distribution.

3.7 Wage effects without ripple effects

In the simple case with no ripple effects, the average wage effect for group g decomposes as

$$d \ln \bar{w}_g = \underbrace{\frac{1}{\lambda} d \ln y}_{\text{productivity effect}} - \underbrace{\frac{1}{\lambda} \delta_g}_{\text{displacement effect}} - \underbrace{\left(\frac{\mu_{A_g}}{\mu_g} - 1 \right) \delta_g}_{\text{rent dissipation}}. \tag{13}$$

Interpretation: The first two terms are standard in competitive task models. The third term is the paper's new channel: if newly automated tasks had above-average rents, exposed workers lose more than the standard displacement effect.

3.8 TFP effects

The TFP effect can be written as

$$d \ln TFP = \sum_g s_g \delta_g \frac{\mu_{A_g}}{\mu_g} \pi_g - \sum_g s_g \left(\frac{\mu_{A_g}}{\mu_g} - 1 \right) \delta_g. \tag{14}$$

Interpretation: The first term is the direct cost-saving gain from automation. The second term is the loss in allocative efficiency from reallocating workers away from high-rent tasks into lower-VMPL jobs. This is why private adoption can overstate social productivity gains.

3.9 Reduced-form wage specification

The group-level reduced form is

$$\Delta Y_g = \beta \,\text{Task displacement}_g + X_g' \gamma + u_g, \tag{15}$$

where Y_g can be average wages, base wages, rents, or a percentile of the within-group wage distribution.

For within-group percentiles, the paper estimates

$$\Delta \ln w_g(p) = \beta(p) \,\text{Task displacement}_g + X_g' \gamma(p) + u_{gp}. \tag{16}$$

Interpretation: The slope $\beta(p)$ tells how a 1 percentage point increase in task displacement changes the wage at percentile p of the within-group distribution. A U-shaped $\beta(p)$ is evidence that automation is removing high-rent jobs rather than simply lowering all wages proportionally.

3.10 Rent-shift specification

The second empirical strategy measures whether exposed groups leave high-rent jobs:

$$\sum_n Rents_{gn} \Delta \ell_{gn} = \beta \,\text{Task displacement}_g + X_g' \gamma + u_g. \tag{17}$$

Interpretation: A negative β means automation pushes group g away from jobs with high rent proxies. This directly tests the mechanism behind the within-group U-shape.

3.11 General equilibrium estimation

The quantitative exercise estimates the rate of rent dissipation ρ and ripple effects using the equations

$$\Delta \ln w_g = \beta_0 - \frac{1}{\lambda} \delta_g + \beta X_g + J_{\Theta,g} \,\text{stack}(\Delta \ln w_j) + u_g, \tag{18}$$

$$\Delta \ln \mu_g = \beta_0^\mu - \rho \delta_g + \beta^\mu X_g^\mu + J_{\mu,g} \,\text{stack}(\Delta \ln w_j) + e_g. \tag{19}$$

Interpretation: This step transforms reduced-form evidence into a general equilibrium decomposition. The propagation matrix spreads base-wage effects across groups, while the rent-impact matrix tracks rent effects from task reallocation.

4 Identification and empirical design

4.1 What identifies the reduced-form relationships

- Identification comes from cross-group differences in exposure to automation-driven task displacement.
- Exposure is determined by baseline industry employment and routine-task specialization, interacted with industry labor-share declines attributed to automation proxies.
- The main specification controls for gender, education, manufacturing exposure, sectoral demand shifts, and sectoral rent shifts.
- Groups are weighted by their 1980 employment shares.

Interpretation: The empirical design is a shift-share style design at the demographic-group level. Groups that started in routine tasks inside automating industries become more exposed.

4.2 Key identifying assumption

The key assumption is that, conditional on controls, groups with higher measured task displacement would not have had systematically lower wage growth or larger within-group compression absent automation.

Interpretation: This is not a randomized experiment. The credibility comes from the detailed exposure construction, the ability to control for sectoral demand and composition changes, and the fact that several distinct predictions line up: average wage losses, within-group compression, and shifts away from rent proxies.

4.3 Separating rents from compensating differentials

- A high wage could be a rent, but it could also compensate workers for bad amenities or risks.
- The paper uses job-quality measures from the 1977 Quality of Employment Survey to purge wages of compensating differentials.
- The U-shaped pattern and rent-shift results remain after this adjustment.

Interpretation: This distinction is central for welfare. Automating dangerous or unpleasant jobs may be efficient; automating rent-paying jobs can reduce allocative efficiency.

4.4 Why the U-shape is important

- A standard skill-biased technology story would usually increase inequality both between and within groups.
- The paper predicts the opposite within exposed groups: upper-middle percentiles lose more than lower percentiles because rents are concentrated in those jobs.
- The U-shape therefore acts as a diagnostic test of rent dissipation.

Interpretation: The U-shape is more than a visual pattern. It is the empirical signature that links the task model to the data.

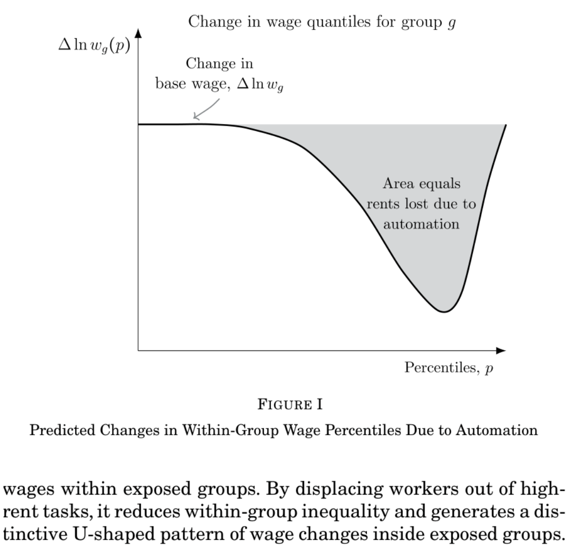
4.5 Limits and threats

- Other shocks could be correlated with automation exposure, including trade, outsourcing, changing unionization, and demand shifts.
- The task-displacement measure relies on assumptions about routine tasks and industry labor-share declines.
- The quantitative exercise assumes cost savings of 30%, common rent dissipation in the baseline, and parameterized ripple matrices.
- The fixed-wedge assumption is a simplification; in reality rents, bargaining, and institutions may change over time.
- The welfare results depend heavily on distinguishing rents from compensating differentials.

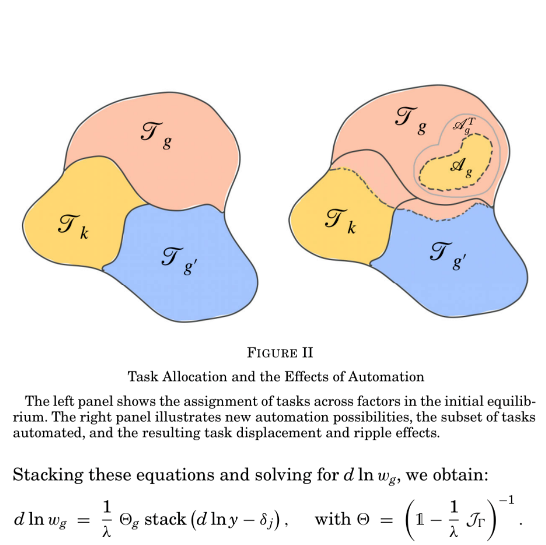
Interpretation: The paper's empirical strategy is strong because multiple mechanisms align, but the quantitative welfare numbers are model-based and should be read as disciplined estimates rather than direct experimental treatment effects.

5 Tables and figures

5.1 Figure I and Figure II: theory of wage compression and ripple effects



(a) Figure I: predicted within-group wage-percentile changes



(b) Figure II: task allocation and automation ripple effects

Read: Figure I:

- The horizontal axis is the percentile within an exposed group; the vertical axis is the wage change.
- The bottom of the distribution experiences the base-wage change.
- Upper-middle percentiles experience larger losses because those workers held rent-paying jobs that were more likely to be automated.
- The highest percentiles can recover toward the base-wage change if some very high-rent tasks are not automatable.

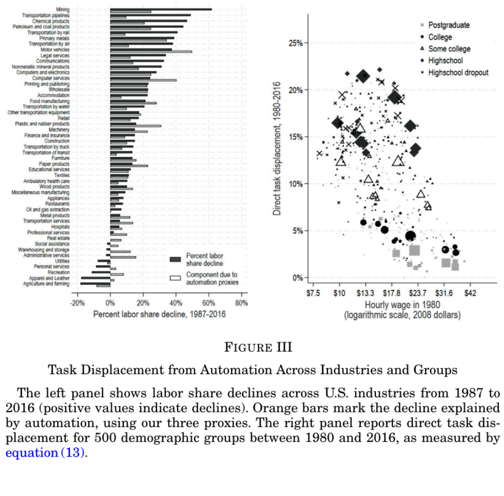
Read: Figure II:

- The left panel shows initial task allocation across groups and capital.
- The right panel shows that new automation opportunities directly displace one group from some tasks.
- Displaced workers compete for nearby marginal tasks, affecting other groups through ripple effects.

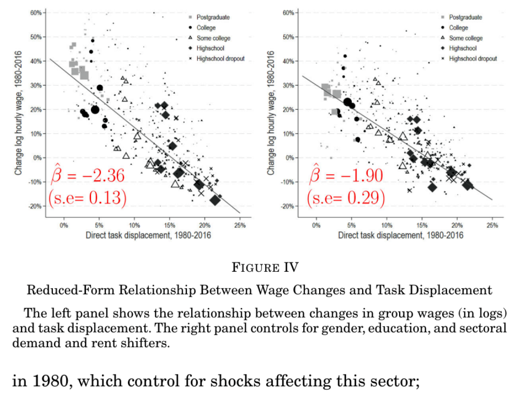
Interpretation: Figure I is the cleanest theoretical prediction: automation can compress wages inside exposed groups. Figure II shows why group-level automation exposure is not only a direct effect; displaced workers alter the task allocation of other groups.

Economic message: The mechanism is not simply "machines replace workers." It is "machines replace the jobs where firms privately save the most wage costs," and those jobs may be high-rent jobs.

5.2 Figure III and Figure IV: measuring exposure and average wage effects



(a) Figure III: task displacement across industries and groups



(b) Figure IV: wage changes and task displacement

Read: Figure III:

- The left panel compares observed labor-share declines across industries with the component attributed to automation proxies.
- The right panel plots task displacement for 500 demographic groups against their baseline hourly wage.
- Middle- and lower-middle-wage groups are most exposed, with task displacement around 15%–20% for many of them.
- Postcollege groups are much less exposed to task displacement.

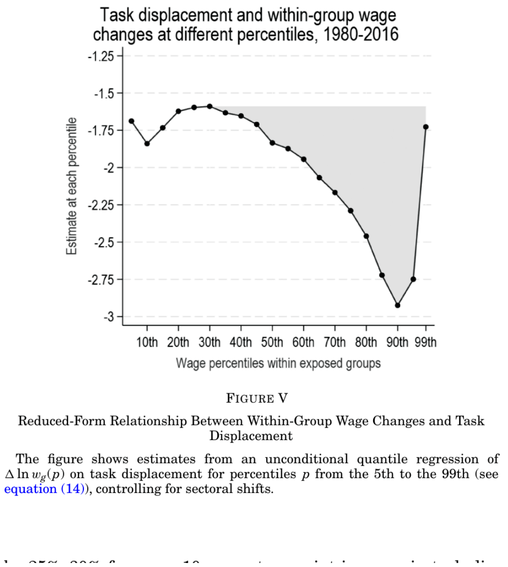
Read: Figure IV:

- The left panel shows a strong negative relationship between task displacement and average group wage growth.
- A 10 percentage point increase in displacement is associated with roughly a 24% decline in group-level relative wages without controls.
- The right panel adds controls and still finds a large negative slope of about -1.9, implying about a 20% wage decline for a 10 percentage point increase in displacement.
- The controlled relationship explains a large share of group wage-change variation.

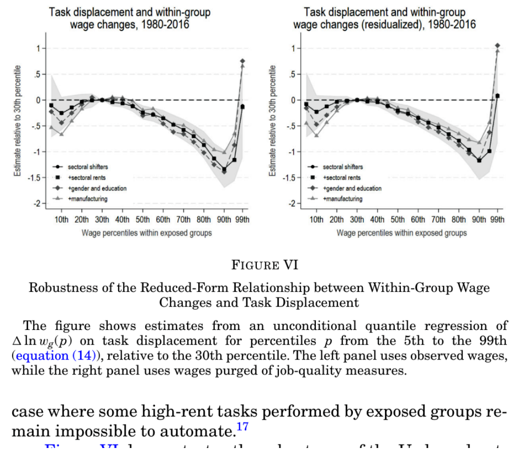
Interpretation: Figure III establishes exposure. Figure IV shows that this exposure predicts wage declines. The concentration of exposure among noncollege and lower-middle-wage groups explains why automation is relevant for rising between-group inequality.

Economic message: The reduced-form slope is the empirical bridge between the theory and U.S. wage changes: groups that lost more tasks to automation also lost more wage growth.

5.3 Figure V and Figure VI: the U-shape in within-group wage changes



(a) Figure V: within-group quantile effects



(b) Figure VI: robustness and residualized wages

Read: Figure V:

- The estimates are effects of task displacement on wage changes at different within-group percentiles.
- Effects at the bottom are negative but relatively flat.
- Effects become much more negative between about the 50th and 95th percentiles.
- The shaded area represents the extra wage decline beyond the base-wage decline, interpreted as rent dissipation.

Read: Figure VI:

- The U-shape remains after adding controls for sectoral rent shifts, gender and education, and manufacturing exposure.
- The right panel uses wages purged of job-quality differences.
- The pattern remains even after trying to remove compensating differentials.

Interpretation: The U-shape is the core rent-dissipation evidence. If automation simply reduced demand for a group, wages would likely shift down more uniformly. The fact that upper-middle percentiles lose more suggests that automation removes jobs that carried rents.

Economic message: This finding reverses a common intuition: automation can increase inequality across groups while decreasing dispersion within groups that are directly exposed.

5.4 Figure VII: decomposing base wages and rents

Read:

- The left panel uses the 30th percentile as the base-wage proxy.
- A 10 percentage point increase in task displacement is associated with about a 15.3% decline in base wages.

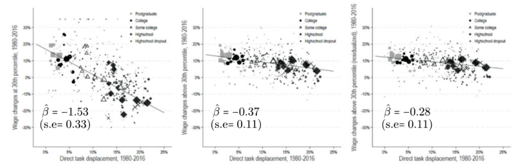


FIGURE VII

Reduced-Form: Changes in Base Wages and Rents from Automation

The left panel plots the relationship between task displacement and changes in base wages, measured by the change at the 30th percentile, $\Delta \ln w_g(30\text{th})$. The middle panel plots the relationship between task displacement and rent losses, measured by the additional wage decline above the 30th percentile, $\Delta \ln \bar{w}_g - \Delta \ln w_g(30\text{th})$. The right panel repeats the exercise but purges wages of job-quality differences. All specifications include our baseline controls: gender and education dummies, sectoral demand and rent shifts, and exposure to manufacturing.

The right-most panel shows estimates of rent dissipation, as in the middle panel, but now also controlling for compensating differentials. Namely, the dependent variable $\Delta \ln \bar{w}_g - \Delta \ln w_g(30\text{th})$

- The middle panel measures the additional decline in average wages above the 30th percentile.
- The implied rent-dissipation rate is about 37%.
- The right panel purges job-quality differences and still implies rent dissipation around 28%.

Interpretation: Figure VII turns the U-shape into a quantitative rent measure. It says that one part of the wage decline is standard displacement, and the additional decline above the 30th percentile is the loss of rents.

Economic message: The paper's key number, a roughly 35% rent-dissipation rate, is not arbitrary. It is anchored in the difference between lower-percentile wage changes and upper-percentile wage losses.

5.5 Figure VIII and Figure IX: direct rent proxies

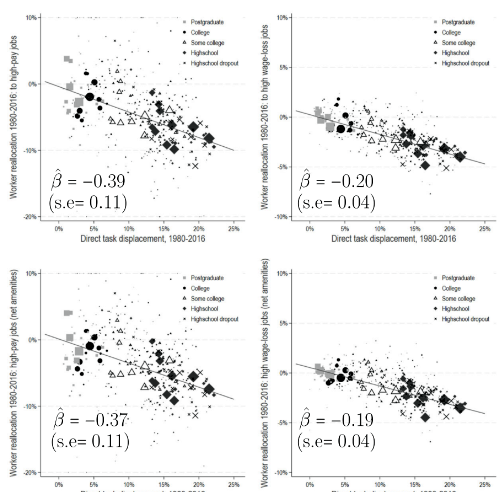


FIGURE VIII

Reduced-Form: Automation and Shifts away from High-Rent Jobs

The figure plots the relationship between automation and worker shifts away from high-rent jobs, measured as $\sum_n \text{Rents}_{gn} \Delta\ell_{gn}$. The top left panel uses industry and occupation wage differentials in 1980 as the proxy for rents. The top right panel uses wage losses after job displacement from the CPS Displaced Worker Supplement. The bottom panels repeat these analyses after purging job-quality differences from the rent proxies. All specifications include baseline controls: gender and education dummies, sectoral demand and rent shifts, and exposure to manufacturing.

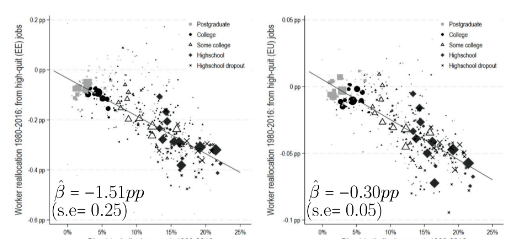


FIGURE IX

Reduced-Form: Automation and Shifts away from High-Rent Jobs (as Revealed by Quits)

The figure plots the relation between automation and worker shifts away from high-rent jobs, measured as $-\sum_n \text{Quit rate}_{gn} \Delta\ell_{gn}$. The left panel presents results using the employment-to-employment monthly transition rate as an inverse proxy for rents (computed from the Basic Monthly CPS). The right panel presents results using the voluntary employment-to-unemployment monthly transition rate as an inverse proxy for rents (computed from the Basic Monthly CPS). All specifications include baseline controls: gender and education dummies, sectoral demand and rent shifts, and exposure to manufacturing.

(a) Figure VIII: shifts away from high-rent jobs using wage and displacement-loss proxies

(b) Figure IX: shifts away from low-quit, high-rent jobs

Read: Figure VIII:

- The top-left panel uses industry-occupation wage differentials as rent proxies.
- A 10 percentage point increase in task displacement is associated with a roughly 3.9% rent decline using this proxy.
- The top-right panel uses wage losses after displacement and implies rent dissipation around 20%.
- The bottom panels repeat the analysis after purging job-quality differences; estimates remain similar.

Read: Figure IX:

- Quit rates are used as inverse rent proxies: workers are less likely to quit attractive rent-paying jobs.
- Automation-exposed groups move toward jobs with higher quit rates.
- Converted into wage-equivalent rents, the estimates imply rent dissipation around 28% using employment-to-employment transitions and around 45% using voluntary employment-to-unemployment transitions.

Interpretation: Figures VIII and IX provide mechanism evidence independent of the quantile U-shape. They show that exposed workers are not merely earning less; they are reallocating away from jobs that look like rent-paying jobs under several rent proxies.

Economic message: The rent-dissipation result is not driven by one measure. Wage differentials, displacement losses, job-quality residuals, and quit behavior all point to the same force: automation destroys access to high-rent employment.

5.6 Table I: general equilibrium quantitative effects

TABLE I

ESTIMATED EFFECTS OF NEW AUTOMATION TECHNOLOGY, 1980–2016.

	Data 1980–2016 (1)	Baseline estimates (2)	Low rent dissipation (20%) (3)	High rent dissipation (50%) (4)	Imposing aggregate elasticity of substitution of 1.6 between groups (5)	Higher elasticity of substitution between industries ($\eta = 0.75$) (6)
Panel A: Wage structure						
Change in average wages, $d \ln \bar{w}_g$ (%)	29.15	0.456	1.91	−0.705	0.497	0.61
Change in base wages, $d \ln w_g$ (%)		4.54	4.54	4.54	4.54	4.54
Change in group rents, $d \ln \mu_g$ (%)		−4.08	−2.63	−5.24	−4.04	−3.92
Wage changes for noncollege men (%)	−6.5	−8.0	−6.8	−9.0	−6.9	−6.5
Base wage changes (%)		−2.4	−3.2	−1.7	−1.3	−0.9
Rent dissipation (%)		−5.6	−3.6	−7.3	−5.6	−5.4
Wage changes for noncollege women (%)	10.6	−2.2	−0.8	−3.3	−1.5	−2.6
Base wage changes (%)		2.4	2.1	2.6	3.0	1.9
Rent dissipation (%)		−4.6	−2.9	−5.9	−4.6	−4.5
Between-group wage changes explained (%)						
Due to changes in industry composition (%)		6.72	7.30	6.26	5.92	-4.76
Adding direct displacement effects (%)		63.49	71.16	57.35	62.68	52.00
Accounting for ripple effects (%)		41.79	46.84	37.75	35.08	34.23
Accounting for rent dissipation (%)		51.87	53.38	50.66	45.39	44.13

TABLE I

CONTINUED

	Data 1980–2016 (1)	Baseline estimates (2)	Low rent dissipation (20%) (3)	High rent dissipation (50%) (4)	Imposing aggregate elasticity of substitution of 1.6 between groups (5)	Higher elasticity of substitution between industries ($\eta = 0.75$) (6)
Panel B: Aggregates						
Change in GDP per capita, $d \ln y$ (%)	70	14.41	15.90	13.22	14.50	14.02
Change in TFP, $d \ln \text{TFP}$ (%)	35	0.304	1.27	−0.47	0.331	0.41
Cost-saving gains (%)		3.02	3.02	3.02	3.02	3.02
Changes in allocative efficiency (%)		−2.72	−1.75	−3.49	−2.69	−2.61
Inefficient rent dissipation (%)		−2.70	−1.73	−3.47	−2.7	−2.7
Change in consumption, $d \ln c$ (%)	70	0.456	1.91	−0.705	0.497	0.61
Change in $\frac{k}{y}$ ratio (%)	30.0	27.9	28.0	27.8	28.0	26.8

Notes. This table summarizes the estimated effects of automation on group wages and aggregates from 1980 to 2016. The estimates are computed using the formulas in Proposition 6. Column (1) reports data for the United States for comparison. The wage data is from the U.S. Census and the ACS. The data on GDP and capital are from the BLS. The data on consumption is from the BEA and the data for TFP is from Fernald (2014). Column (2) provides our baseline estimates. Column 3 assumes a lower rate of rent dissipation of 20%. Column (4) assumes a higher rate of rent dissipation of 50%. Column (5) assumes $\theta = 1.6 - \lambda$ and sets $\theta_{\text{job}} = \theta_{\text{edu-age}} = 0$, imposing an aggregate elasticity of substitution between worker groups of 1.6. Column (6) uses the baseline specification from column (2) but sets the elasticity of substitution between industries η to 0.75.

Read: Panel A:

- The baseline estimate of the change in average wages is only 0.456%, despite a 4.54% base-wage gain.
- Group rents fall by 4.08%, nearly wiping out the base-wage gain.
- Noncollege men are predicted to lose 8.0% in wages, compared with a 6.5% decline in the data.
- Noncollege women are predicted to lose 2.2%, despite a 10.6% gain in the data, indicating that automation was only one of several forces affecting their wages.
- The contribution of automation to between-group wage changes rises from 41.79% after ripple effects to 51.87% after adding rent dissipation.

Read: Panel B:

- Automation raises GDP by 14.41% in the baseline calculation.
- The direct cost-saving gain to TFP is 3.02%.
- Allocative efficiency worsens by 2.72%, almost entirely due to inefficient rent dissipation.
- Net TFP rises by only 0.304%, and consumption rises by only 0.456%.
- Under low rent dissipation, net TFP is 1.27%; under high rent dissipation, net TFP is -0.47%.

Interpretation: Table I shows that automation can increase measured output while producing small consumption and TFP gains. The large GDP effect reflects investment and task reallocation; the small TFP effect reflects inefficient rent dissipation.

Economic message: The table separates private and social returns. Firms automate because it saves wages, but if those wages include rents, the social benefit can be much smaller than the private cost saving.

Policy interpretation: The policy implication is not that all automation should be blocked. It is that labor-market distortions change the social return to automation. Policies that reduce inefficient rents, improve mobility, or support task creation could raise the social value of new technologies.

5.7 Figure X and Figure XI: model decomposition and fit

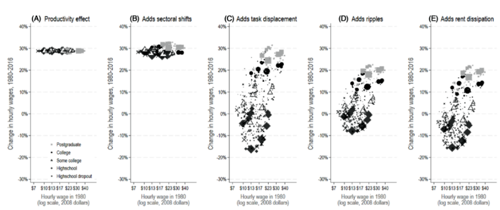


FIGURE X

Wage Effects of Automation

The figure shows estimates of automation's impact on between-group wage changes through different channels, beginning with the productivity effect in Panel A and adding components cumulatively until Panel E. In all panels, the horizontal axis sorts groups by their hourly wage in 1980.

(a) Figure X: wage effects by channel

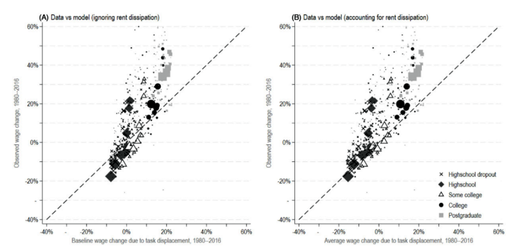


FIGURE XI

Wage Effects of Automation: Model Versus Data

The left panel compares predicted wage changes, excluding rent dissipation, to observed changes from 1980–2016. The right one includes rent dissipation and shows the full model-predicted wage changes against data.

accounting for rent dissipation, automation cannot explain the real wage declines experienced by many low-education groups between 1980 and 2016. Once rent dissipation is included, our esti-

(b) Figure XI: model versus data

Read: Figure X:

- Panel A shows a common productivity gain for all groups.
- Panel B adds sectoral shifts, which create only modest dispersion.
- Panel C adds direct displacement and generates large wage losses for exposed groups.
- Panel D adds ripple effects, which reduce dispersion because displaced groups compete for marginal tasks and shift the incidence to others.
- Panel E adds rent dissipation, which again increases inequality because exposed groups bear the full loss of rents.

Read: Figure XI:

- The left panel compares actual wage changes with model predictions that omit rent dissipation.
- The right panel includes rent dissipation.
- Including rent dissipation moves predictions closer to the observed wage stagnation and declines for low-education groups.
- The paper reports that base-wage effects alone explain about 42% of observed between-group wage changes, while the full model explains about 52%.

Interpretation: Figure X is the paper's decomposition map. It shows why competitive forces alone are not enough: productivity effects are broad, direct displacement is unequalizing, ripple effects are partially equalizing, and rent dissipation is an additional unequalizing force. Figure XI then shows why rent dissipation matters empirically: without it, the model misses too much of the real wage decline among exposed groups.

Economic message: The model's final prediction is not just that automation hurts some groups. It is that automation reorders the wage structure through a sequence of mechanisms: productivity gains, task displacement, general equilibrium competition, and rent destruction.

6 Short summary

- The paper studies automation in a task economy with labor-market rents.
- Rents are wage wedges that make firms pay more than workers' opportunity costs in some tasks.
- Because high-rent tasks have high private labor costs, firms have stronger incentives to automate them.
- Automation therefore dissipates rents by pushing workers out of high-rent jobs.
- This amplifies wage losses for exposed groups beyond the standard task-displacement effect.
- It also predicts within-group wage compression and a U-shaped profile of wage changes across percentiles.
- The empirical analysis uses 500 U.S. demographic groups from 1980 to 2016.
- Task displacement is measured using baseline industry exposure, routine-task specialization, and industry labor-share declines attributed to robots, software, and dedicated machinery.
- Groups more exposed to automation experience larger average wage declines.
- Exposed groups also show the predicted U-shaped within-group wage pattern.
- Rent proxies based on wage differentials, displacement losses, and quit rates confirm that automation shifts workers away from high-rent jobs.
- The estimated rent-dissipation rate is around 35%.
- In the quantitative exercise, automation explains about 52% of the rise in between-group wage inequality since 1980.
- Rent dissipation explains about one-fifth of this automation effect.
- Direct cost savings from automation raise TFP by about 3%, but rent dissipation offsets most of this productivity gain.
- The baseline net TFP gain is only about 0.3%, and the consumption gain is about 0.46%.
- The paper's main lesson is that the social value of automation depends on the labor-market distortions in the jobs being automated.

7 Conclusion

7.1 Core conclusion

Acemoglu and Restrepo show that automation interacts with labor-market rents in a way that changes both distributional and aggregate outcomes. In competitive models, automation has a clear productivity benefit, even when it reduces wages for exposed groups. In this paper, the same technology can have a weaker or even negative social productivity effect because firms automate tasks where wages are privately high due to rents, not necessarily where labor has low social value.

7.2 Economic intuition

The paper's intuition is simple but powerful. A firm compares the cost of labor in a task to the cost of capital or software. If the task pays a rent, the wage bill is high, and automation looks privately attractive. But the planner compares the cost of capital to the worker's outside option, not to the rent-inflated wage. Therefore, a technology that saves firms money may partly work by destroying worker rents rather than by improving productive efficiency.

This distinction is why the productivity gains are much smaller than the private cost savings. The machine saves the firm the rent-paying wage, but society also loses the surplus that workers used to receive from that high-rent match. When displaced workers move into lower-VMPL jobs, allocative efficiency deteriorates.

7.3 Labor-market interpretation

The results imply that automation should be studied together with institutions such as unions, wage norms, licensing, bargaining, efficiency wages, monopsony, and internal labor markets. These institutions affect which jobs pay rents, and therefore which jobs firms want to automate first. The technology frontier alone does not determine automation. Adoption also depends on the wage structure created by labor-market frictions.

This is why the paper's empirical focus on within-group wage percentiles is so useful. Average wages alone can show that exposed groups lose. The within-group distribution reveals *which jobs* within the group disappear. A U-shape indicates that rent-paying jobs in the upper-middle part of the distribution are being removed.

7.4 Inequality implications

The paper adds a new layer to the automation-and-inequality debate. Automation can raise between-group inequality by reducing demand for noncollege and routine-task groups, but it can also reduce within-group dispersion among exposed groups by removing rent-paying jobs. Thus, inequality need not rise fractally at every level. The same shock can widen gaps between demographic groups while compressing wages within the groups most exposed to the shock.

This matters for interpreting observed wage trends. If within-group inequality among some noncollege groups falls or stagnates, that does not necessarily mean technology was irrelevant. It may mean that technology eliminated the high-rent jobs that used to generate within-group wage dispersion.

7.5 Productivity and welfare implications

The paper challenges the view that automation is always aggregate-productivity enhancing. In the model, productivity gains have two components. The direct component is the cost saving from automating tasks. The second component is the allocative-efficiency effect from moving labor across tasks. If automation removes workers from high-rent, high-VMPL jobs, the allocative-efficiency effect can be negative and large.

The quantitative results suggest that this issue is economically large. Direct cost savings are estimated to raise TFP by about 3%, but inefficient rent dissipation offsets 60%–90% of these gains. The baseline TFP contribution is only around 0.3%, and the consumption gain is only around 0.46%. This does not imply automation is unimportant. It means that much of automation's private value may come from redistribution and rent destruction rather than pure productive efficiency.

7.6 Policy implications

The policy message is subtle. A policy that simply preserves all high-rent jobs can maintain inefficient labor wedges and block genuinely productivity-enhancing technologies. But a policy that blindly subsidizes automation can encourage firms to replace workers in jobs where the social value of labor remains high. The more constructive policy lesson is to reduce the mismatch between private and social automation incentives.

Possible directions include:

- improving worker mobility and retraining so displaced workers do not move into low-VMPL jobs;
- encouraging technologies that create new tasks for workers rather than only replacing existing tasks;
- scrutinizing tax and subsidy systems that may already favor capital over labor;
- designing labor-market institutions that preserve useful risk-sharing or bargaining benefits without creating excessive wedges that induce inefficient automation;
- using place-based or group-specific adjustment policies when automation destroys high-rent local or occupational jobs.

7.7 Managerial and firm-level implications

For firms, the paper clarifies that automation incentives are shaped by wage premia. Managers may identify the highest-wage jobs as the most attractive targets for automation, but high wages can reflect rents, match quality, or institutional premia rather than low social productivity. Therefore, cost-based automation decisions can overemphasize rent savings and underemphasize longer-term organizational or worker-skill losses.

7.8 Research agenda

The paper opens several empirical questions. Matched employer-employee data could identify whether firms automate specifically in jobs with high establishment wage premia. Technology-adoption data could test whether firms with larger labor wedges adopt more automation. More detailed data on amenities and job quality could separate rents from compensating differentials more cleanly. Finally, similar methods could be applied to trade, outsourcing, immigration, or AI adoption to see whether these shocks also dissipate rents.

7.9 Final takeaway

The main lesson is that automation is not merely a technology shock. It is also a rent-reallocation shock. When labor markets are imperfect, automation can destroy high-rent jobs, amplify wage losses for exposed workers, increase between-group inequality, and offset much of the productivity gain that would arise in a competitive economy. The welfare evaluation of automation therefore depends not only on what machines can do, but also on why the jobs they replace were highly paid.